EDGE INTELLIGENCE ASSESSMENT – 2

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1. MNIST

ipynb file : [dr0b1n/Lab_4]

This project uses a simple neural network to recognize handwritten digits from the MNIST dataset, this model was then saved using Pickle and its performance was evaluated.

1. Data Jar:

The data jar contains everything the model learns from, here it corresponds to digit images from the MNIST dataset. These images are used to teach the model what the numbers 0 to 9 look like.

2. Task Jar:

The task is to look at an image and recognize which digit it represents. ie: the model is trained to recognize and label numbers correctly.

3. Model Jar:

Consists of the neural network that we use to look at the images, to find patterns that help tell one number apart from another. As

we train it on more data, it slowly becomes better at making the right guesses.

4. Loss Jar:

The loss jar measures how wrong the model's guesses are. If the model predicts the wrong digit, the loss becomes larger which helps the system understand when it needs to improve.

5. Learning Jar:

The learning jar is about how the model changes itself to get better. Using the loss values, the model adjusts its internal settings so it can make fewer mistakes next time. Over many rounds of training, this improves the model's predictions.

6. Accuracy Jar:

The accuracy jar shows how well the model is performing, by checking accuracy on new images we can see how well the model recognizes numbers.

```python
import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow import keras
from tensorflow.keras.datasets import mnist

(x_full_train, y_full_train), (x_test, y_test) = mnist.load_data()

train_size = 50000

x_train, y_train = x_full_train[:train_size], y_full_train[:train_size]
x_val, y_val = x_full_train[train_size:], y_full_train[train_size:]

x_train = x_train.astype('float32') / 255.0
x_val = x_val.astype('float32') / 255.0

x_train = x_train.reshape(-1, 28, 28, 1)
x_val = x_val.reshape(-1, 28, 28, 1)

model = tf.keras.Sequential([
    tf.keras.layers.Input(shape=(28, 28, 1)),
    tf.keras.layers.Conv2D(32, (3, 3), activation='relu'),
    tf.keras.layers.MaxPooling2D((2, 2)),
    tf.keras.layers.Flatten(),
    tf.keras.layers.Dense(10, activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

history = model.fit(x_train, y_train, epochs=6, validation_data=(x_val, y_val), verbose=1)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Epoch 1/6
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 27s 16ms/step - accuracy: 0.8724 - loss: 0.4615 - val_accuracy: 0.9726 - val_loss: 0.1100
Epoch 2/6
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 40s 16ms/step - accuracy: 0.9711 - loss: 0.1008 - val_accuracy: 0.9761 - val_loss: 0.0820
Epoch 3/6
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 26s 16ms/step - accuracy: 0.9801 - loss: 0.0669 - val_accuracy: 0.9772 - val_loss: 0.0782
Epoch 4/6
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 40s 16ms/step - accuracy: 0.9831 - loss: 0.0555 - val_accuracy: 0.9824 - val_loss: 0.0645
Epoch 5/6
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 26s 17ms/step - accuracy: 0.9863 - loss: 0.0458 - val_accuracy: 0.9831 - val_loss: 0.0641
Epoch 6/6
1563/1563 ━━━━━━━━━━━━━━━━━━━━ 25s 16ms/step - accuracy: 0.9888 - loss: 0.0368 - val_accuracy: 0.9820 - val_loss: 0.0636
```

```
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
flatten (Flatten)	(None, 5408)	0
dense (Dense)	(None, 10)	54,090

Total params: 163,232 (637.63 KB)
Trainable params: 54,410 (212.54 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 108,822 (425.09 KB)

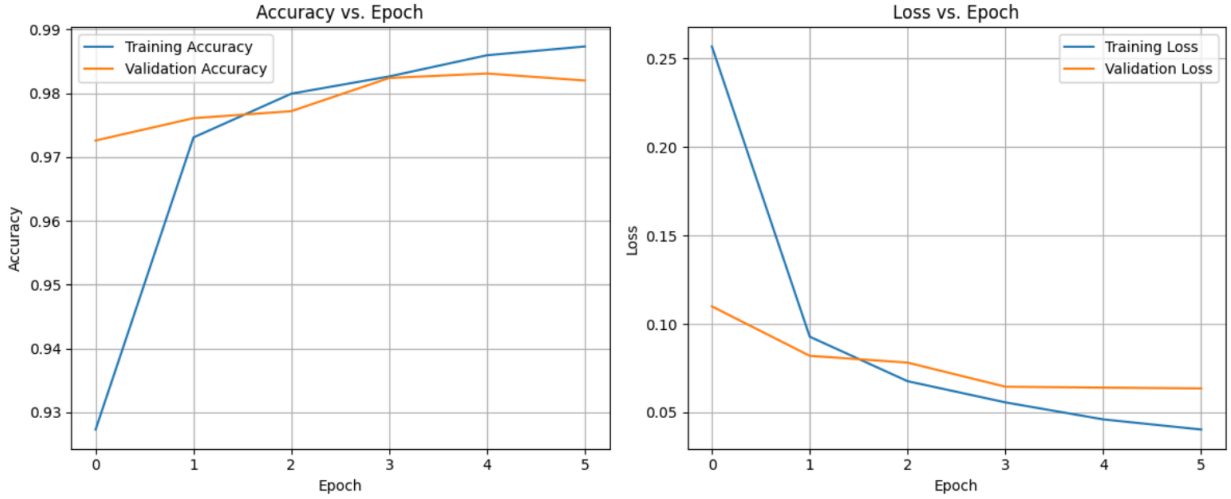
```python
import pickle

with open('pickle_model.pkl', 'wb') as file:
    pickle.dump(model, file)

print('Model saved as pickle_model.pkl')
```

```
Model saved as pickle_model.pkl
```

```python
from google.colab import files

files.download('pickle_model.pkl')
```

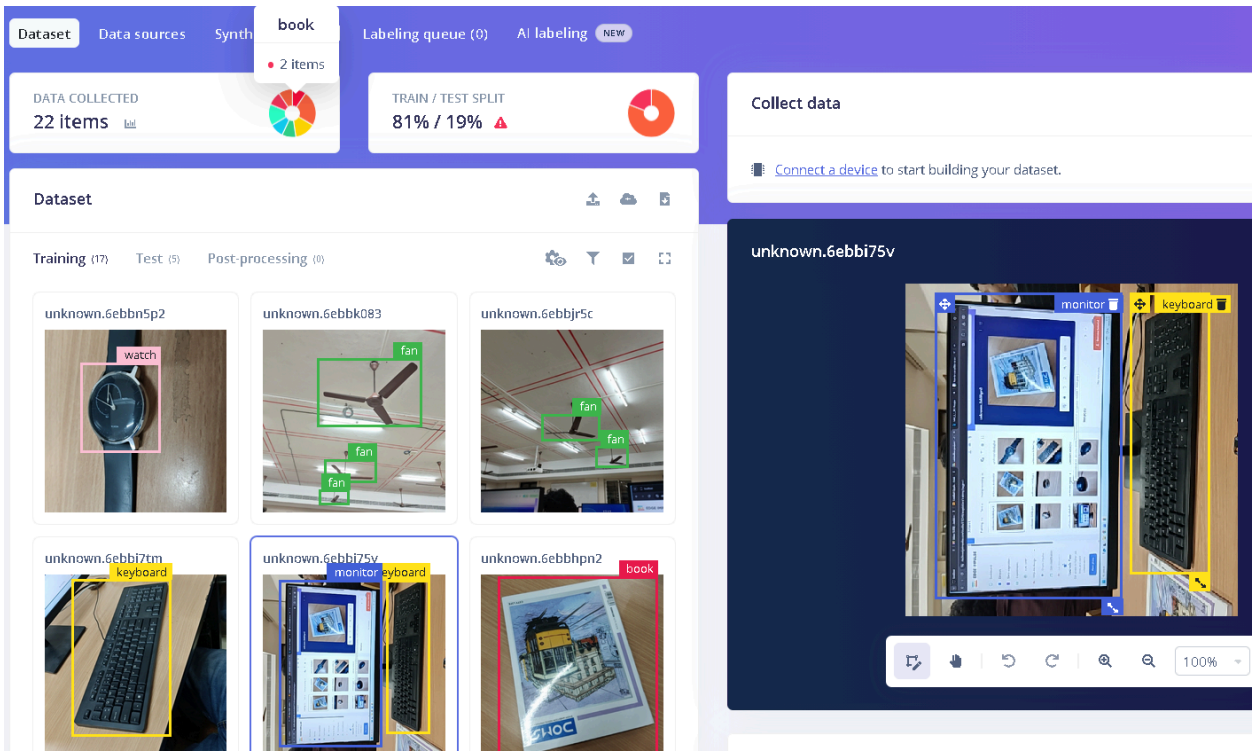
```python
with open('pickle_model.pkl', 'rb') as file:
    pickled_model = pickle.load(file)
```

```python
import matplotlib.pyplot as plt

plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Accuracy vs. Epoch')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.grid(True)

plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Loss vs. Epoch')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
```



2. Edge Impulse

Training data:



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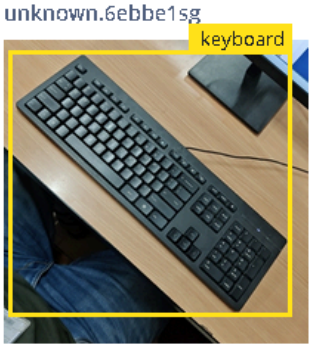
unknown.6ebbgha5

unknown.6ebbepst

unknown.6ebbegsb

unknown.6ebbe1sg



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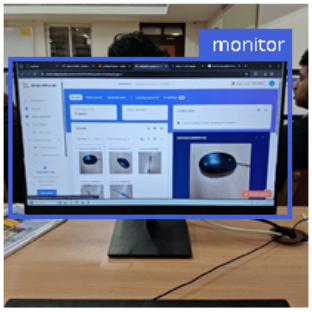

unknown.6ebbd1mb

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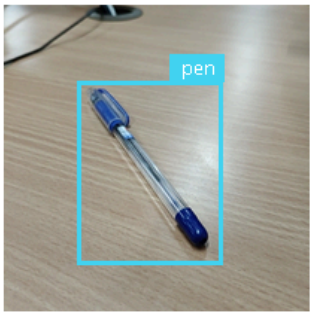
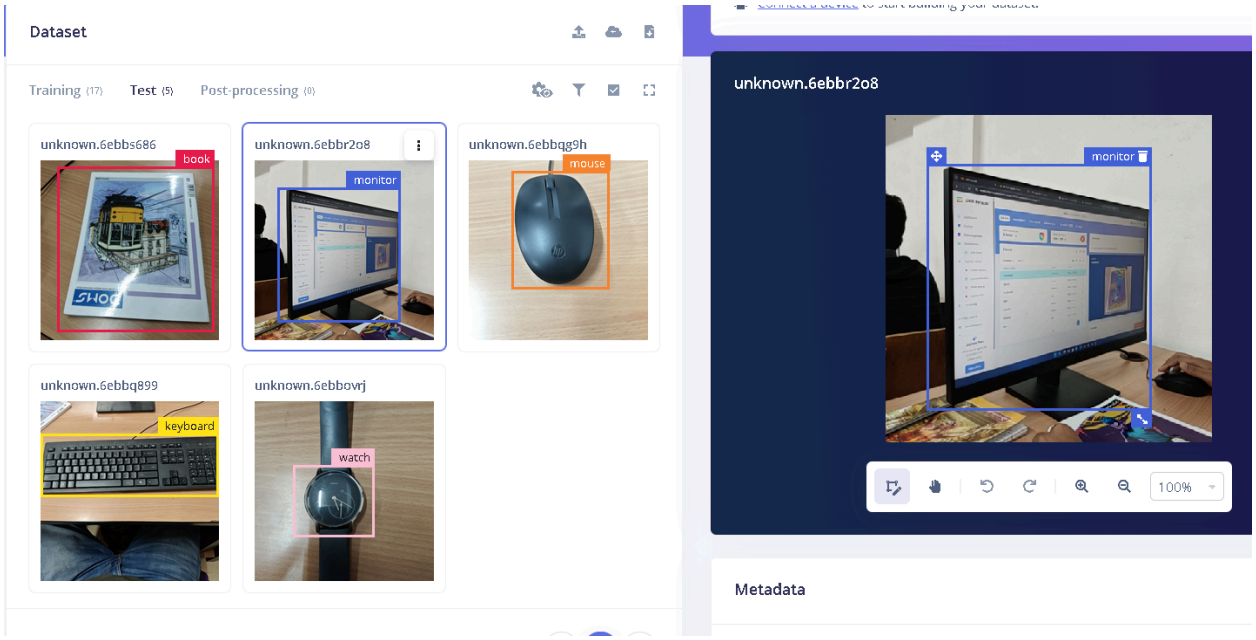


unknown.6ebb5nqf

test data:


Dataset
Training (17)
Test (5)
Post-processing (0)
unknown.6ebbs686
book
unknown.6ebbr2o8
monitor
unknown.6ebbqg9h
mouse
unknown.6ebbq899
keyboard
unknown.6ebbovrj
watch
unknown.6ebbr2o8
monitor
100%
Metadata